

Question # 1 of 10 ( Start time: 08:10:51 PM, 07 November 2017 )

Which one is not an initial segment of  $\mathbb{N}$

Select the correct option

|                                  |                                                                                   |
|----------------------------------|-----------------------------------------------------------------------------------|
| <input type="radio"/>            | $\{1, 2, 3, \dots, 30\}.$<br><i>AHMAD MSC(MATH)</i><br><i>Areeb2113@gmail.com</i> |
| <input type="radio"/>            | $\{1, 2, 3, \dots, 100, 101, 102, 103\}.$                                         |
| <input type="radio"/>            | $\{1, 2, 3\}.$                                                                    |
| <input checked="" type="radio"/> | $\{2, 4, 6\}.$                                                                    |

The supremum of the set  $B = \{p \in \mathbb{Q} : p^2 < 3\}$  is

▶ Select the correct option



$\sqrt{2}$ .

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$\sqrt{3}$ .



10.



19.

Question # 3 of 10 ( Start time: 08:12:24 PM, 07 November 2017 )

The union of set of rational numbers and the set of irrational numbers is

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Select the correct option



The set  $\{n \in \mathbb{Q} : x^7 - 10x + 30 = 0\}$ .



The set of integers.



The set of natural numbers.



$\mathbb{R}$ , the set of real numbers.

## Question # 4 of 10 ( Start time: 08:13:00 PM, 07 November 2017 )

The supremum of the set  $E_2 = \{1/n : n \in \mathbb{N}\}$  is

Select the correct option

|                                  |                                                           |
|----------------------------------|-----------------------------------------------------------|
| <input type="radio"/>            | 5<br><i>AHMAD MSC(MATH)</i><br><i>Areeb2113@gmail.com</i> |
| <input checked="" type="radio"/> | 1                                                         |
| <input type="radio"/>            | 7                                                         |
| <input type="radio"/>            | 4                                                         |

Question # 5 of 10 ( Start time: 08:13:37 PM, 07 November 2017 )

Identify the false statement(s)

Select the correct option



The set of natural numbers is not complete.

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The set of rational numbers is complete



The set of rational numbers is an infinite set.



The set of real numbers is complete.

Question # 6 of 10 ( Start time: 08:14:46 PM, 07 November 2017 )

The infimum of set of integers is

Select the correct option



$-\infty$ .

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$\infty$ .



$-100$ .



$10$ .



Click t

The statement(s) true about the real number  $4\sqrt{10}$  is(are)

Select the correct option



is a rational number.

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an irrational number.



the solution of  $x^2 - 2x + 3$ .



the solution of  $x^2 - 4 = 0$ .

Which statement(s) is(are) true

Select the correct option



The set of even numbers is defined as  $\{2k - 1 : k \in \mathbb{N}\}$ .

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The set of odd numbers is defined as  $\{2k : k \in \mathbb{N}\}$ .



Both set of even numbers and set of odd numbers are countable.



The set of even numbers is a finite set.



Which set is dense in the of real number

Select the correct option



The set of integers.

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The set of rational numbers.



The set  $[0, 1]$ .



All of the above.

Which one is an irrational number

Select the correct option



$$\frac{22}{7}$$

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$$\pi$$



$$\frac{p+q}{pq}$$
, where  $p, q$  are nonnegative integers.



$$\sqrt{s}$$
, where  $s$  is a perfect square.

Question # 1 of 10 ( Start time: 10:45:18 PM, 16 May 2018 )

Which statement is true about the sets  $\mathbb{N}$  and  $\mathbb{Q}$

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Select the correct option

☐  $\mathbb{N} = \mathbb{Q}$

☐  $\mathbb{Q} \subset \mathbb{N}$ .

☒  $\mathbb{N} \subset \mathbb{Q}$ .

☐ None of the above.

Click t

Which statement(s) is(are) true

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Select the correct option



The set of even numbers is defined as  $\{2k - 1 : k \in \mathbb{N}\}$ .



The set of odd numbers is defined as  $\{2k : k \in \mathbb{N}\}$ .



Both set of even numbers and set of odd numbers are countable.



The set of even numbers is a finite set.

The supremum of the set  $E_2 = \{1/n : n \in \mathbb{N}\}$  is

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Select the correct option

|                                  |   |
|----------------------------------|---|
| <input type="radio"/>            | 5 |
| <input checked="" type="radio"/> | 1 |
| <input type="radio"/>            | 7 |
| <input type="radio"/>            | 4 |

Click

The supremum of the set  $B = \{p \in \mathbb{Q} : p^2 < 3\}$  is

Select the correct option



$\sqrt{2}$ .

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*Areeb2113@gmail.com*



$\sqrt{3}$ .



10.



19.

The union of set of rational numbers and the set of irrational numbers is

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Select the correct option



The set  $\{n \in \mathbb{Q} : x^7 - 10x + 30 = 0\}$ .



The set of integers.



The set of natural numbers.



$\mathbb{R}$ , the set of real numbers.

The infimum of the set  $E_2 = \{1/n : n \in \mathbb{N}\}$  is

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Select the correct option



-1.



2.



-3.



0.



The statement(s) true about the real number  $4\sqrt{10}$  is(are)

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Select the correct option



is a rational number.



an irrational number.



the solution of  $x^2 - 2x + 3$ .



the solution of  $x^2 - 4 = 0$ .

The intersection of set of odd numbers and the set of even numbers is

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Select the correct option

|                                  |                              |
|----------------------------------|------------------------------|
| <input type="radio"/>            | The set of rational numbers. |
| <input checked="" type="radio"/> | The empty set.               |
| <input type="radio"/>            | $\{1, 2, 3, 4\}$ .           |
| <input type="radio"/>            | The set of real numbers.     |

Question # 10 of 10 ( Start time: 10:49:12 PM, 16 May 2018 )

Identify the false statement(s)

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Select the correct option



The set of natural numbers is not complete.



The set of rational numbers is complete



The set of rational numbers is an infinite set.



The set of real numbers is complete.



Which statement(s) is(are) true about the following sequence  $s_0 = 1$  and  $s_n = 1 - e^{-s_{n-1}}$

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Select the correct option



$0 < s_n \leq 1$  for all  $n$ .



$s_{n+1} - s_n = -(e^{-s_n} - e^{-s_{n-1}})$ .



Sequence is convergent.



All of the above.

quiz.vu.edu.pk/QuizQuestion.asp

4



MC170403773: HUMA JAVED

Time Left 86 sec(s)

MTH621: Quiz 1

Quiz Start Time: 08:30 PM, 14 May 2018

Question # 4 of 10 ( Start time: 08:31:44 PM, 14 May 2018 )

Total Marks: 1

Which statement is true about the sets  $\mathbb{N}$  and  $\mathbb{Q}$ 

Select the correct option

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Insert Math Equations

|                       |                                 |
|-----------------------|---------------------------------|
| <input type="radio"/> | $\mathbb{N} = \mathbb{Q}$       |
| <input type="radio"/> | $\mathbb{Q} \subset \mathbb{N}$ |
| <input type="radio"/> | $\mathbb{N} \subset \mathbb{Q}$ |
| <input type="radio"/> | None of the above.              |

100% correct answers (0/0)

Question # 2 of 10 ( Start time: 02:07:17 PM, 20 November 2017 )

If the sequence is increasing, then it

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▶ Select the correct option



converges to its supremum.



diverges.



may converges to its supremum.



is bounded.

Which of the following given sets is compact subset of  $\mathbb{R}$  .

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Select the correct option



$(1,5]$



$\{1,2,4\}$



$[0,1]$



$(2,3)$

Identify the identical sequences.

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Select the correct option



$$\left\{ \frac{1}{n^2-2} \right\}_3^\infty \text{ and } \left\{ \frac{1}{n} \right\}_1^\infty.$$



$$\left\{ \frac{1}{n} \right\}_3^\infty \text{ and } \left\{ \frac{1}{n} \right\}_1^\infty.$$



$$\left\{ \frac{1}{n-2} \right\}_3^\infty \text{ and } \left\{ \frac{1}{n^2} \right\}_1^\infty.$$



$$\left\{ \frac{1}{n-2} \right\}_3^\infty \text{ and } \left\{ \frac{1}{n} \right\}_1^\infty.$$



Which statement(s) is(are) true about the following sequence  $s_0 = 1$  and  $s_n = 1 - e^{-s_{n-1}}$

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Select the correct option

|                                  |                                                |
|----------------------------------|------------------------------------------------|
| <input type="radio"/>            | $0 < s_n \leq 1$ for all $n$ .                 |
| <input type="radio"/>            | $s_{n+1} - s_n = -(e^{-s_n} - e^{-s_{n-1}})$ . |
| <input type="radio"/>            | Sequence is convergent.                        |
| <input checked="" type="radio"/> | All of the above.                              |

$\limsup_{n \rightarrow \infty} s_n = \infty$  if

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Select the correct option



$\{s_n\}$  is not bounded above.



$\{s_n\}$  has unique limit.



$\{s_n\}$  is convergent.



$\{s_n\}$  has limit inferior.



Type here to search



The series  $\sum_{n=0}^{\infty} r^n$ ,  $-1 < r < 1$ , is

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Select the correct option



divergent.



convergent and converges to  $\frac{1}{1+r^2}$ .



convergent and converges to 5.



convergent and converges to  $\frac{1}{1-r}$ .



The set of terms of  $\{s_{n_k}\}$  is contained in the set of terms of  $\{s_n\}$  implies

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Select the correct option



$$\sup\{s_n\} = \sup\{s_{n_k}\}.$$



$$\sup\{s_n\} \leq \sup\{s_{n_k}\}.$$



$$\sup\{s_n\} \geq \sup\{s_{n_k}\}.$$



$\{s_n\}$  is convergent.

Every Cauchy sequence has a

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▶ Select the correct option



convergent subsequence.



increasing subsequence.



decreasing subsequence.



positive subsequence.

Question # 10 of 10 ( Start time: 02:15:53 PM, 28 November 2017 )

The limit of the sequence

$$s_n = \frac{1}{n} + \frac{2(1 + 3/n)}{1 + 1/n}.$$

Select the correct option

|                                  |   |                                                      |
|----------------------------------|---|------------------------------------------------------|
| <input type="radio"/>            | 1 | <i>AHMAD MSC(MATH)</i><br><i>Areeb2113@gmail.com</i> |
| <input type="radio"/>            | 4 |                                                      |
| <input type="radio"/>            | 3 |                                                      |
| <input checked="" type="radio"/> | 2 |                                                      |

If  $p > 0$  and  $\alpha$  is real, then value of the

$$\lim_{n \rightarrow \infty} \frac{n^\alpha}{(1+p)^n},$$

Select the correct option



$\infty$ .

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3.



2.



0.



Type here to search



The value of the

$$\lim_{n \rightarrow \infty} \frac{1}{n^p}$$

Select the correct option



$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0 \text{ for } p > 1.$$

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$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 3 \text{ for all } p.$$



$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 1 \text{ for all } p.$$



$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 1 \text{ for } p > 1.$$



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Question # 3 of 10 ( Start time: 09:47:35 PM, 30 November 2017 )

If the sequence is convergent then

Select the correct option



it has two limits.

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it is bounded.



it is bounded above but may not be bounded below.



it is bounded below but may not be bounded above.



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Which statement(s) is(are) true about the following sequence  $s_0 = 1$  and  $s_n = 1 - e^{-s_{n-1}}$

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Select the correct option

|                                  |                                                |
|----------------------------------|------------------------------------------------|
| <input type="radio"/>            | $0 < s_n \leq 1$ for all $n$ .                 |
| <input type="radio"/>            | $s_{n+1} - s_n = -(e^{-s_n} - e^{-s_{n-1}})$ . |
| <input type="radio"/>            | Sequence is convergent.                        |
| <input checked="" type="radio"/> | All of the above.                              |



Type here to search



The series  $\sum_{n=0}^{\infty} r^n$ ,  $-1 < r < 1$ , is

Select the correct option



divergent.

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*Areeb2113@gmail.com*



convergent and converges to  $\frac{1}{1+r^2}$ .



convergent and converges to 5.



convergent and converges to  $\frac{1}{1-r}$ .



Type here to search



Which sequence is bounded

Select the correct option

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$$s_n = [1 + (-1)^n]n.$$



$$\{(-1)^n\}.$$



$$\{n\}.$$



$$\{\frac{1}{n} + n^2\}.$$



Type here to search



The value of the  $\lim_{n \rightarrow \infty} \frac{1}{n^p}$  is

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Select the correct option



$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0 \text{ for } p > 0.$$



$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 3 \text{ for all } p.$$



$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 1 \text{ for all } p.$$



$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 1 \text{ for all } p > 1.$$



Type here to search



Every Cauchy sequence has a

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Select the correct option



convergent subsequence.



increasing subsequence.



decreasing subsequence.



positive subsequence.



Type here to search



Question 7 of 15 (Start time: 00:00:11, 00 November 2017)

Which statement(s) is(are) true for the following sequence  $\lim_{n \rightarrow \infty} \sqrt[n]{n}$

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Select the correct option

|                                  |                              |
|----------------------------------|------------------------------|
| <input type="radio"/>            | The sequence is oscillatory. |
| <input checked="" type="radio"/> | The sequence converges to 1. |
| <input type="radio"/>            | The sequence converges to 0. |
| <input type="radio"/>            | The sequence is divergent to |

Which of the following given sets is compact subset of  $\mathbb{R}$  .

Select the correct option



$(1,5]$

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$\{1,2,4\}$



$[0,1]$



$(2,3)$



Type here to search





The infimum of set of integers is:

Select the correct option

|                                  |           |
|----------------------------------|-----------|
| <input checked="" type="radio"/> | $-\infty$ |
| <input type="radio"/>            | $\infty$  |
| <input type="radio"/>            | $-100$    |
| <input type="radio"/>            | $10$      |

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The supremum of a set is

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Select the correct option



not unique.



the greatest upper bound of that set.



least upper bound.



None of the above.

The infimum of the set  $B = \{p \in \mathbb{Q} : p^2 > 2\}$  is

Select the correct option

|                       |               |
|-----------------------|---------------|
| <input type="radio"/> | $\sqrt{2}$ .  |
| <input type="radio"/> | $\sqrt{10}$ . |
| <input type="radio"/> | $\sqrt{5}$ .  |
| <input type="radio"/> | 1.            |

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The infimum of the set  $B = \{p \in \mathbb{Q} : p^2 > 2\}$  is

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Select the correct option.

|                       |             |
|-----------------------|-------------|
| <input type="radio"/> | $\sqrt{2}$  |
| <input type="radio"/> | $\sqrt{10}$ |
| <input type="radio"/> | $\sqrt{5}$  |
| <input type="radio"/> | 3           |

Click to Save Answer

The infimum of the set  $E_2 = \{1/n : n \in \mathbb{N}\}$  is

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Select the correct option

|                                  |     |
|----------------------------------|-----|
| <input type="radio"/>            | -1. |
| <input type="radio"/>            | 2.  |
| <input type="radio"/>            | -3. |
| <input checked="" type="radio"/> | 0.  |

Identify the countable set

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Select the correct option

|                       |                                  |
|-----------------------|----------------------------------|
| <input type="radio"/> | The set of irrational numbers.   |
| <input type="radio"/> | The interval $[0, 1]$ .          |
| <input type="radio"/> | $\mathbb{N} \times \mathbb{N}$ . |
| <input type="radio"/> | The interval $[1, 2]$ .          |

Question # 3 of 10 ( Start time: 08:47:49 PM, 14 May 2018 )

Identify the false statement(s)

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Select the correct option



The set of natural numbers is not complete.



The set of rational numbers is complete



The set of rational numbers is an infinite set.



The set of real numbers is complete.

[Click to Save Answer & Move](#)

The intersection of set of odd numbers and the set of even numbers is

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Select the correct option

|                                  |                              |
|----------------------------------|------------------------------|
| <input type="radio"/>            | The set of rational numbers. |
| <input checked="" type="radio"/> | The empty set.               |
| <input type="radio"/>            | $\{1, 2, 3, 4\}$ .           |
| <input type="radio"/>            | The set of real numbers.     |





MC170400773: HUMA JAVED

MTH621:Quiz 1

Question # 5 of 10 ( Start time: 08:32:36 PM, 14 May 2018 )

The union of set of rational numbers and the set of irrational numbers is

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Select the correct option

|                       |                                                       |
|-----------------------|-------------------------------------------------------|
| <input type="radio"/> | The set $\{n \in \mathbb{Q} : x^7 - 10x + 30 = 0\}$ . |
| <input type="radio"/> | The set of integers.                                  |
| <input type="radio"/> | The set of natural numbers.                           |
| <input type="radio"/> | $\mathbb{R}$ , the set of real numbers.               |

The supremum of the set  $B = \{p \in \mathbb{Q} : p^2 < 3\}$  is

Select the correct option

|                       |              |                                                    |
|-----------------------|--------------|----------------------------------------------------|
| <input type="radio"/> | $\sqrt{2}$ . | <i>AHMAD MSC(MA)</i><br><i>Areeb2113@gmail.com</i> |
| <input type="radio"/> | $\sqrt{3}$ . |                                                    |
| <input type="radio"/> | 10.          |                                                    |
| <input type="radio"/> | 19.          |                                                    |

## Question # 7 of 10 ( Start time: 08:34:21 PM, 14 May 2018 )

The sum of a rational number and an irrational number is

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Select the correct option

|                       |                                                     |
|-----------------------|-----------------------------------------------------|
| <input type="radio"/> | may be a rational depends on the numbers.           |
| <input type="radio"/> | always an irrational number.                        |
| <input type="radio"/> | always a rational number.                           |
| <input type="radio"/> | may be an irrational number depends on the numbers. |

## Question # 7 of 10 ( Start time: 08:34:21 PM, 14 May 2018 )

The sum of a rational number and an irrational number is

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Select the correct option

|                       |                                                     |
|-----------------------|-----------------------------------------------------|
| <input type="radio"/> | may be a rational depends on the numbers.           |
| <input type="radio"/> | always an irrational number.                        |
| <input type="radio"/> | always a rational number.                           |
| <input type="radio"/> | may be an irrational number depends on the numbers. |

An upper bound for the set  $A = \{p \in \mathbb{Q} : p^2 < 2\}$  is

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Select the correct option

|                       |    |
|-----------------------|----|
| <input type="radio"/> | -1 |
| <input type="radio"/> | 0  |
| <input type="radio"/> | 4  |
| <input type="radio"/> | -4 |

ANS

Question # 1 of 10 ( Start time: 08:30:27 PM, 14 May 2018 )

The supremum of the set  $S_2 = \{1/n : n \in \mathbb{N}\}$  is

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Select the correct option

|                       |   |
|-----------------------|---|
| <input type="radio"/> | 5 |
| <input type="radio"/> | 1 |
| <input type="radio"/> | 7 |
| <input type="radio"/> | 4 |

ANSWER

Question # 3 of 10 ( Start time: 08:31:17 PM, 14 May 2018 )

The infimum of the set  $E_2 = \{1/n : n \in \mathbb{N}\}$  is*AHMAD MSC(MATH)**Areeb2113@gmail.com*

Select the correct option

|                                  |     |
|----------------------------------|-----|
| <input type="radio"/>            | -1. |
| <input type="radio"/>            | 2.  |
| <input type="radio"/>            | -3. |
| <input checked="" type="radio"/> | 0.  |

Question # 2 of 10 ( Start time: 08:30:45 PM, 14 May 2018 )

An upper bound for the set  $A = \{p \in \mathbb{Q} : p^2 < 2\}$  is

Select the correct option

|                                  |     |                                                      |
|----------------------------------|-----|------------------------------------------------------|
| <input type="radio"/>            | -1. | <i>AHMAD MSC(MATH)</i><br><i>Areeb2113@gmail.com</i> |
| <input type="radio"/>            | 0.  |                                                      |
| <input checked="" type="radio"/> | 4.  |                                                      |
| <input type="radio"/>            | -4. |                                                      |



The value of the  $\lim_{n \rightarrow \infty} \frac{1}{n^p}$  is

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Select the correct option



$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0 \text{ for } p > 0.$$



$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 3 \text{ for all } p.$$



$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 1 \text{ for all } p.$$



$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 1 \text{ for all } p > 1.$$



Type here to search



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(22)

### ABP's Test

Series  $\sum a_n b_n$  Converges.

if  $a_{n+1} \leq a_n$  For  $n \geq k$

$\lim_{n \rightarrow \infty} a_n = 0$  and

$$|b_k + b_{k+1} + \dots + b_n| \leq M, n \geq k$$

Q#1:- Cauchy convergence criterion - (5)

Q#2:-  $\sqrt{8}$  is always irrational number (5)

Q#3:- Show that  $(n - n^2)$  converges or diverges. (2)

Q#3:- Give an example of subsequence which does not converges but its subsequence does converges: (3)

Q#4:-  $S \supseteq (-\infty, -1) \cup (1, 2) \cup (3)$  give all limit points of  $S$ . (3)

Q#5:-  $\{S_n\} = \left\{ \frac{1}{n} \right\} = \left\{ 1 + \frac{1}{2}, \frac{1}{3}, \dots \right\}$   
write subsequence :- (2)

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Mth621 my today's paper

1) state convergence sequence

2) is irrational is countable

3) check convergence of series

Long

4)  $\sqrt{8}$  is irrational prove

5) prove "e"

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Mth621

Mcqs conceptual thy

Infimum definition

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Series power

Bounded sequence thrm tha...

Underroot 8 rational hai yh prove  
krna tha 5 marks

RABEES TEST 5 marks

12:23 PM

Mth621

Write any subsequence of sequence  $1/n^2$   
2marks

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Define infimum 2marks

Discuss series convergent or divergent  $(n/2 + 1/n)$   
3marks

Cauchy's criterion state and proof 5

Aik theorem tha koi yad nai wo b 5 ka  
tha

12:34 PM



MTH 621

AHMAD MSC(MATH)  
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Mcqs were v-tough and  
only two mcqs from quiz.

- 1) Sequence Given thi uski Limit  
find krni thi- (2)
- 2) Power Series def. (2)
- 3) Prove that Convergent Series  
is bounded. (3)
- 4) Define Limit inferior for  $S_n$ - (3)
- 5) Find the approximation of  $e$ - (5)
- 6) Principle of mathematical  
induction se 1 question- (5)